# Yield Curve Models and Data

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## Nominal Yield Curve

The U.S. Treasury regularly issues Treasury securities (depending on their maturity, these securities are known as bills, notes, or bonds). These securities are promises to repay the principal at maturity, and, in the case of notes and bonds, to make periodic interest payments. When principal and any interest payments are specified as fixed dollar amounts at the time of issuance, the security is known as a "nominal" Treasury security (compare with Treasury inflation-protected securities (TIPS)). Since Treasury securities are backed by the full faith and credit of the U.S. government, the returns investors can earn on them are often used as a "risk-free" benchmark in finance research and investment practice.

**Related Content**

FEDS working paper: "The U.S. Treasury Yield Curve: 1961 to the Present"

Investors can trade Treasury securities freely between issuance and maturity. As the market price of Treasury securities varies over time, so does their implied yield—their return relative to their price. At any given time, there is a wide range of Treasury securities with different maturities outstanding. Market forces tend to ensure that the yields on securities with similar maturities are not dramatically different from each other. This feature makes it possible to summarize the information contained in the cross section of market-implied yields by a smooth curve of yield as a function of maturity—the yield curve.

One popular yield curve specification, the Svensson model, stipulates that the shape of the yield curve on any given date can be adequately captured by a set of six parameters.[1] The values of these parameters can be estimated by minimizing the discrepancy between the fitted Svensson yield curve and observed market yields. This Svensson model is used to fit daily yield curves for the period since 1980. Before 1980, the Nelson-Siegel model—a model with fewer parameters—was used to fit the yield curve, as there were not enough Treasury securities to fit the Svensson model.

This page provides daily estimated nominal yield curve parameters, and smoothed yields on hypothetical Treasury securities that can be easily compared across maturities and over time, from 1961 to the present.[2]

**Data:** CSV | HTML

### Q&A

1. **Does the daily yield curve estimation use all existing nominal Treasury securities?**

   No. Treasury bills and Treasury floating rate notes (FRNs) are not used; only the coupon securities (Treasury notes and bonds) are used. Furthermore, on-the-run and first-off-the-run Treasury notes and bonds are not used, as these securities often trade at a premium to other Treasury securities due to their greater liquidity and their frequent specialness in the repo market. In short, the data provided here represent an off-the-run nominal yield curve based on Treasury coupon securities.

2. **Are the data provided here the same as the data originally released with the Gurkaynak, Sack, and Wright (2006) paper?**

   No. The current vintage data are generally pretty close to the original Gurkaynak, Sack, and Wright data, but they are not identical, as small modifications have been made over time in the way these models are implemented. Further modifications could be made in the future, so the future vintage data could also differ a bit from the current vintage.

3. **How often are the data updated?**

   We generally try to post the updated series once per week; typically, updated series data will be posted on Tuesday for the period up to the Friday of the previous week.

4. **Is this model an official Board statistical release?**

   No, this model is a staff research product and not an official statistical release. Accordingly, it is subject to delay, revision, or methodological changes without advance notice.

   Model disclaimer

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1. See Lars E. O. Svensson (1994), "Estimating and Interpreting Forward Rates: Sweden 1992-4," National Bureau of Economic Research Working Paper #4871. Return to text

2. For details on the methodology and data, see Refet S. Gürkaynak, Brian Sack, and Jonathan H. Wright (2006), "The U.S. Treasury Yield Curve: 1961 to the Present," Finance and Economics Discussion Series 2006-28, also published as Refet S. Gürkaynak, Brian Sack, and Jonathan H. Wright (2007), Journal of Monetary Economics, vol 54, pp2291-2304. Return to text

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